

RESEARCH ARTICLE

Phenology-Trait Relationships across Different Scales and Organizational Levels

The performance of growing degree day models to predict spring phenology of herbaceous species depends on the species' temporal niche

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Abstract

1. The concept of growing degree days (GDDs) is commonly used to predict phenological events in plants, assuming that plants develop proportionally to the accumulated temperature. Two species-specific parameters, T_{Base} and t_0 (minimum temperature above which and start date when GDDs begin to accumulate), are considered for the calculation. However, species-specific optimised thresholds of wild herbaceous species remain sparse, and therefore the reliability of the models is questionable.
2. By employing several modelling approaches using phenological records of leaf unfolding and flowering onset of 87 wild herbaceous species collected in six European botanical gardens between 2019 and 2024, we assessed the reliability of GDD models across a diverse array of species. We further examined whether thresholds of T_{Base} and t_0 for calculating GDD can be optimised for a large set of species and for single species. We aimed to estimate and evaluate these thresholds and the reliability of GDD models using species' temporal niche and bud traits to see whether for specific groups of species, specific GDD models work better.
3. Our analyses revealed that GDD models for leaf unfolding and flowering onset performed better than the null model (i.e. mean date across years and species) for 84% and 70% of the species, respectively. Our results showed that species with intermediate temporal niches were less dependent on the selection of T_{Base} and t_0 . Overall, we found better performance of the models using a T_{Base} around 4°C for most of the species. By considering optimised thresholds, we found that predictions of leaf unfolding dates were more accurate in early-growing species, and regarding the start date for temperature accumulation, we found that larger

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